

RFID-BASED SECURITY SYSTEM

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A radio-frequency identification (RFID) based access-control system allows only authorised persons to enter a particular area of an establishment. The authorised persons are provided with unique tags, using which they can access that area.

The system is based on microcontroller AT89C52 and comprises an RFID module, an LCD module for displaying the status and a relay for opening the door. Fig. 1 shows a user trying to open the door by placing an RFID tag near the RFID reader.

Radio-frequency identification

You might be familiar with RFID systems as seen in access control, contactless payment systems, product

tracking and inventory control, etc. Basically, an RFID system consists of three components: an antenna or coil, a transceiver (with decoder) and a transponder (RF tag) electronically programmed with unique information.

Fig. 2 shows a typical RFID system. In every RFID system, the transponder tags contain unique identifying information. This information can be as little as a single binary bit or a large array of bits representing such things as an identity code, personal medical



information or literally any type of information that can be stored in digital binary format.

The RFID transceiver communicates with a passive tag. Passive tags have no power source of their own and instead derive power from the incident electromagnetic field. Commonly, at the heart of each tag is a microchip. When the tag enters the generated RF field, it is able to draw enough power from the field to access its internal memory and transmit its stored information. When the transponder tag draws power in this way, the resultant interaction of the RF fields causes the voltage at the transceiver antenna to drop in value. This effect is utilised by the tag to communicate its information to the reader. The tag is able to control the amount of power drawn from the field and by doing so it can modulate the voltage sensed at the transceiver according to the bit pattern it wishes to transmit.

Antenna. Fig. 3 shows the internal diagram of a typical RFID antenna. An RFID antenna consists of a coil with one or more windings and a matching network. It radiates the electromag-

PARTS LIST

Semiconductors:

IC1	- AT89C52 microcontroller
IC2	- 7812, 12V regulator
IC3	- 7805, 5V regulator
IC4	- RFID reader module
T1	- BC558 pnp transistor
T2	- BC547 npn transistor
D1-D5	- 1N4007 rectifier diode
LED1-LED3	- 5mm LED
LCD	- 16-character, 2-line

Resistors (all 1/4-watt, $\pm 5\%$ carbon unless stated otherwise):

R1	- 10-kilo-ohm
R2, R3	- 330-ohm
R4, R5	- 1-kilo-ohm
R6	- 33-ohm

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2, C3	- 0.1 μ F ceramic disk
C4	- 10 μ F, 16V electrolytic
C4, C5	- 22pF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 15V, 500mA secondary transformer
S1	- Push-to-on switch
X _{cr}	- 11.0592MHz crystal
RL1	- 12V, 1C/O relay
PZ1	- Piezobuzzer
	- RFID tag
	- 12V DC door lock actuator



Fig. 1: A user is trying to open the door by placing an RFID tag near the RFID reader

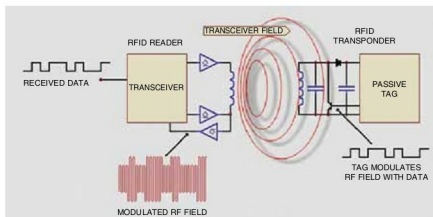


Fig. 2: A typical RFID system